Introduction to stochastic inventory theory

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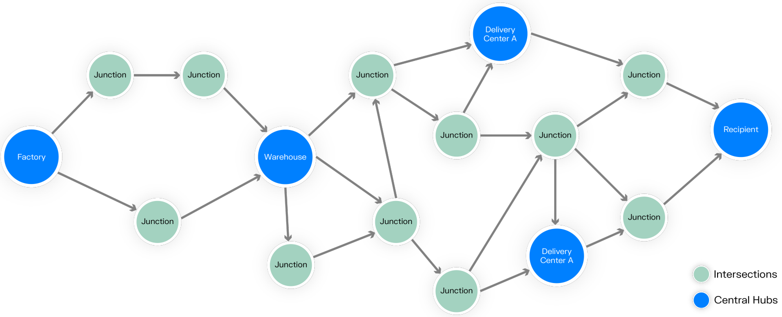
April 2026

Problem formulation:



- We focus on single item (SKU) and single warehouse.

Economic order quantity (EOQ) models

- Constant rate of demand $D\ \left[\frac{\text{units}}{\text{day}}\right]$.
- Shortages are not allowed.
- Cost consists of
  - Transportation costs given by constant C_T [euro],
  - Individual unit cost $C_U\ \left[\frac{\text{euro}}{\text{unit}}\right]$,
  - Holding costs $C_H\ \left[\frac{\text{euro}}{\text{unit} \cdot \text{day}}\right]$.
- Determine order quantity Q which minimizes costs.

Economic order quantity (EOQ) model

- For some order quantity Q denote the corresponding time period t, then $Q = D \cdot t$, and costs are given by:

$$C(Q) := C_T + C_U \cdot Q + C_H \cdot \frac{t \cdot Q}{2}$$

We minimize daily costs $\frac{C(Q)}{t}$:

$$\frac{C(Q)}{t} := \frac{C_T}{t} + \frac{C_U \cdot Q}{t} + C_H \cdot \frac{Q}{2} = \frac{C_T \cdot D}{Q} + C_U \cdot D + C_H \cdot \frac{Q}{2}$$

$$\left(\frac{C(Q)}{t}\right)'_Q = -\frac{C_T \cdot D}{Q^2} + C_H \cdot \frac{1}{2}$$

and the minimum:

$$Q^* = \sqrt{\frac{2 \cdot C_T \cdot D}{C_H}}$$

is called *economic order quantity*.

The single period (newsvendor) model

- Random demand $\mathscr{D} > 0$ $\left[\text{units}\right]$ with CDF Φ.
- Cost consists of
  - Individual unit cost C_U $\left[\frac{\text{euro}}{\text{unit}}\right]$,
  - Holding costs C_H $\left[\frac{\text{euro}}{\text{unit}}\right]$ (for unsold units),
  - Penalty for unsatisfied demand C_P $\left[\frac{\text{euro}}{\text{unit}}\right]$,
  - Transportation costs are set to zero for simplicity.
- Determine order quantity y which minimizes costs.
- $C_P > C_U > -C_H$.

The single period (newsvendor) model

- For some order quantity y denote the corresponding expected costs

$$C(y) = C_U \cdot y + C_H \int_0^y (y - \xi)\, d\Phi_\xi + C_P \int_y^\infty (\xi - y)\, d\Phi_\xi$$

note that

$$\mu = \mathbb{E}(\mathscr{D}) = \int_0^\infty \xi\, d\Phi_\xi, \quad \Phi(y) = \int_0^y d\Phi_\xi, \quad \Phi(\infty) = 1$$

and (y is a constant parameter):

$$y = \mu + (y - \mu) = \mu + \int_0^y (y - \xi)\, d\Phi_\xi - \int_y^\infty (\xi - y)\, d\Phi_\xi$$

then

$$C(y) = C_U \cdot \mu + C_U (y - \mu) + C_H \Big(\ldots \Big) + C_P \Big(\ldots \Big) =$$

$$C_U \cdot \mu + (C_H + C_U) \Big(\ldots \Big) + (C_P - C_U) \Big(\ldots \Big)$$

The single period (newsvendor) model

Similarly using

$$\int_0^\infty (y-\xi)\,d\Phi_\xi = \int_0^y (y-\xi)\,d\Phi_\xi + \int_y^\infty (y-\xi)\,d\Phi_\xi = y-\mu$$

finally get

$$C(y) = C_U \cdot \mu + (C_H + C_U)\Big((y-\mu) + \int_y^\infty (\xi - y)\,d\Phi_\xi\Big) + (C_P - C_U)\int_y^\infty (\xi - y)\,d\Phi_\xi$$

or, equvivalently

$$C(y) = C_U \cdot \mu + (C_H + C_U)(y-\mu) + (C_P + C_H)\int_y^\infty (\xi - y)\,d\Phi_\xi$$

The single period (newsvendor) model

Minimizing:

$$(C(y))'_y = C_H + C_U - (C_P + C_H)(1 - \Phi(y))$$
$$1 - \Phi(y^*) = \frac{C_H + C_U}{C_P + C_H}$$
$$\Phi(y^*) = \frac{C_P - C_U}{C_P + C_H} = \mathrm{SL}$$

where the last variable is called *service level* (SL) and

$$y^* = \Phi^{-1}(\mathrm{SL}) = \Phi^{-1}\Big(\frac{C_P - C_U}{C_P + C_H}\Big).$$

- Service level is a probability of not stocking out.

The single period (newsvendor) model

- $C_P > C_U > -C_H$

$$\text{SL} = \frac{C_P - C_U}{C_P + C_H} = \frac{\text{unit underage cost}}{\text{unit overage cost } + \text{ unit underage cost}}$$

- Example 1: Machine part in warehouse
  - $C_U = 100\,€$, price $110\,€$ then unit underage cost $= 10\,€$.
  - We expect to sell it in the next period, only spend $1\,€$ on storage. $C_H = -99 < 0$ and unit overage cost $= 1\,€$.
  - $\text{SL} = 10/11 \approx 91\%$.

- Example 2: Icecream in supermarket
  - $C_U = 1\,€$, price $2\,€$, then unit underage cost $= 1\,€$.
  - After the period end the product spoils, dispose of it for free. $C_H = 0$ and unit overage cost $= 1\,€$
  - $\text{SL} = 1/2 = 50\%$.

(S, s) model

- Introduce *lead-time* LT > 0 [days], time it takes for order from the supplier to reach us.
- Daily sales are random variables

  $$\mathscr{D}_i$$

  We denote the sum of daily demand over the lead time

  $$\mathscr{D} = \mathscr{D}_1 + \mathscr{D}_2 + \cdots + \mathscr{D}_{\mathrm{LT}-1} + \mathscr{D}_{\mathrm{LT}}.$$

  For big LT take $\mathscr{D} \sim \mathscr{N}(\mu, \sigma)$ (*central limit theorem*).
- We need to compose an optimal ordering strategy.

(S, s) model

- To achieve service level SL we shall have at least s items in stock when we order, i.e.

$$\text{Probability}\big(\mathscr{D} < s\big) = \text{SL}$$

  where s for normally distributed $\mathscr{D} \sim \mathscr{N}(\mu, \sigma)$ is given by

$$s = \mu + \sigma \cdot \Phi^{-1}(\text{SL})$$

  where Φ^{-1} is inverse CDF of standard normal distribution.

- Optimal order size is Q^* is economic order quantity (EOQ). Define

$$S = s + Q^*$$

(S, s) model

- We denote *on-hand inventory* by

$$\mathrm{OI}$$

  and *order position*

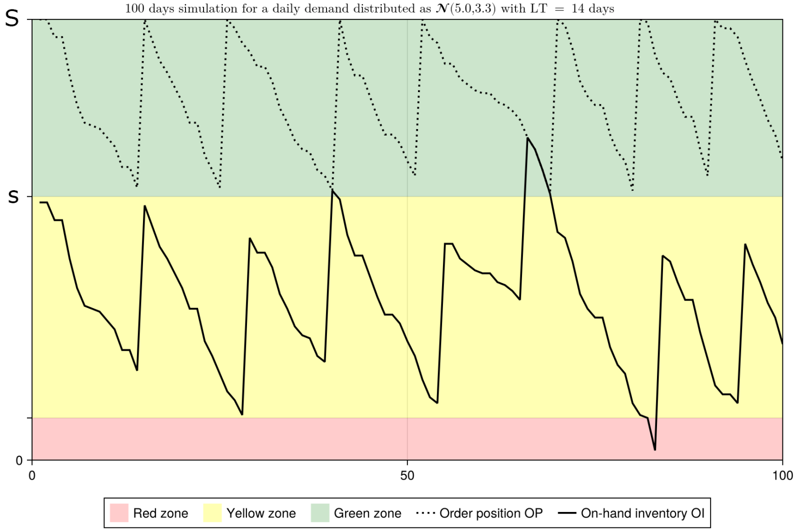
$$\mathrm{OP} = \mathrm{OI} + \text{Items in delivery from the supplier.}$$

- We must maintain $\mathrm{OP} \geq s$ and order Q^*, i.e. always

$$S \geq \mathrm{OP} \geq s$$

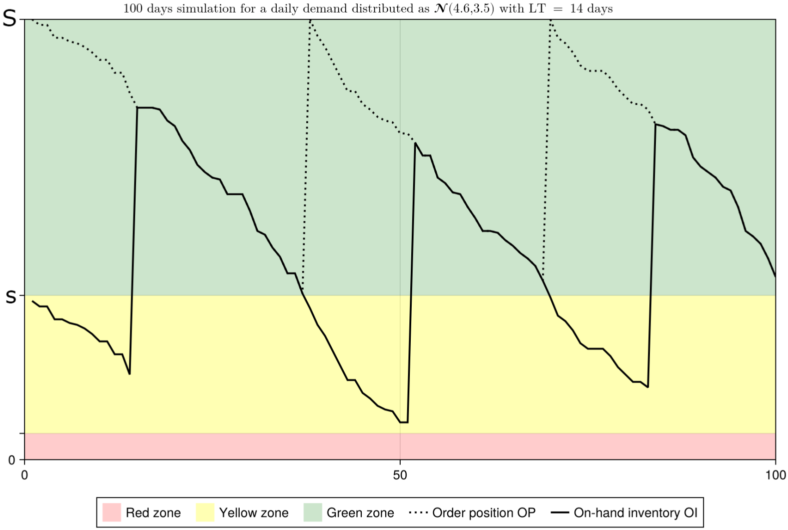
- This is practically (S, s) model for inventory management.

(S, s) model



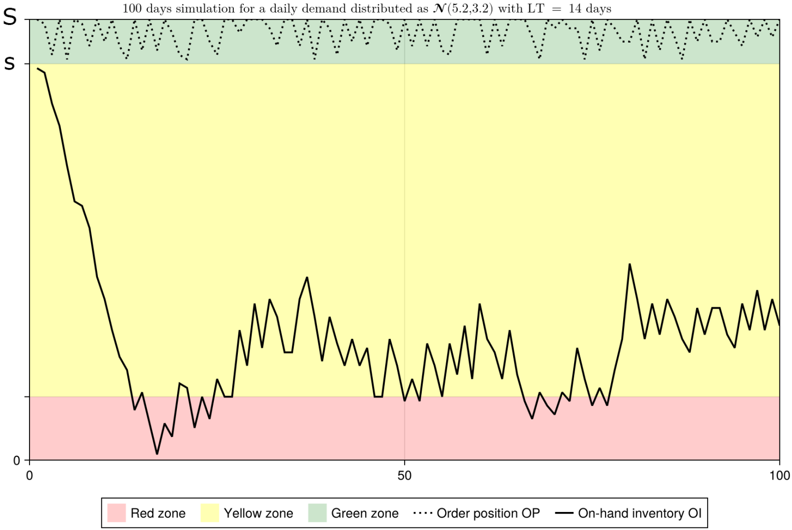
100 days simulation for a daily demand distributed as $\mathcal{N}(5.0,3.3)$ with LT = 14 days
S
s
0
0
50
100
Red zone
Yellow zone
Green zone
Order position OP
On-hand inventory OI

(S, s) model



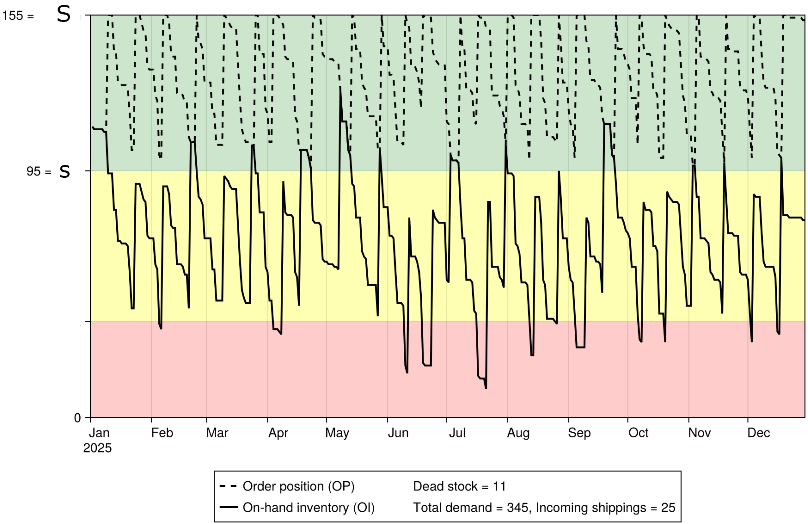
100 days simulation for a daily demand distributed as $\mathcal{N}(4.6,3.5)$ with LT = 14 days
S
s
0
0
50
100
Red zone
Yellow zone
Green zone
Order position OP
On-hand inventory OI

(S, s) model



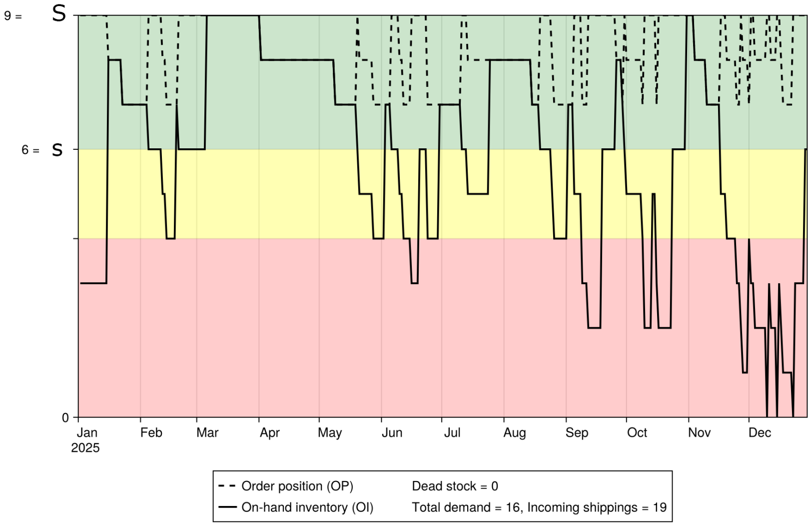
100 days simulation for a daily demand distributed as $\mathcal{N}(5.2,3.2)$ with LT = 14 days
S
s
0
0
50
100
Red zone
Yellow zone
Green zone
Order position OP
On-hand inventory OI

(S, s) model



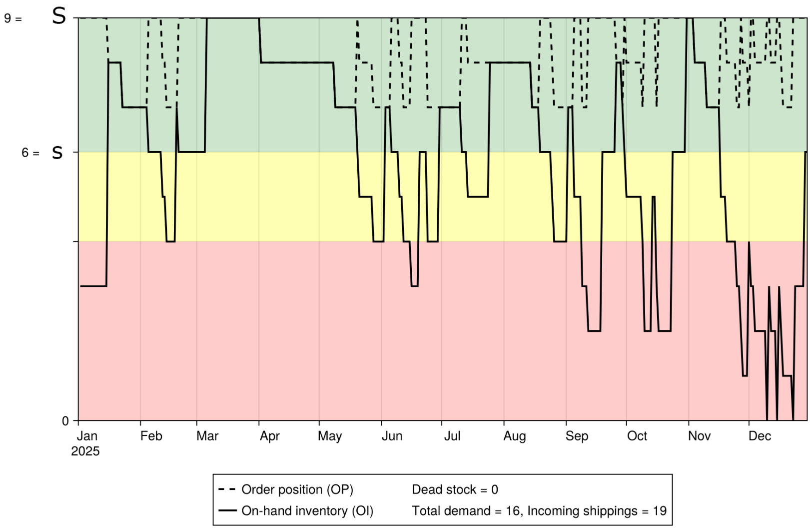
155 =
S
95 =
s
0
Jan
2025
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Order position (OP)
On-hand inventory (OI)
Dead stock = 11
Total demand = 345, Incoming shippings = 25

(S, s) model



9 =
S
6 =
s
0
Jan
2025
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Order position (OP)
On-hand inventory (OI)
Dead stock = 0
Total demand = 16, Incoming shippings = 19

(S, s) model



9 =
S
6 =
s
0
Jan
2025
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Order position (OP)
On-hand inventory (OI)
Dead stock = 0
Total demand = 16, Incoming shippings = 19

References

Thank you for your attention!

Source:

- Evan L. Porteus
  Handbooks in Operations Research and Management Science, Chapter 12 Stochastic inventory theory
  Volume 2, 1990, Pages 605-652